

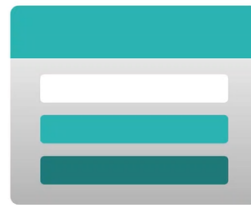
Cloud Computing 101- part 4

Azure Storage, Migration and Identity

Before you can store anything in Azure, you need a container that holds it all together. That container is called a Storage Account. Think of it like opening a bank account before you can deposit money. It comes with a unique name that anyone on the internet can use to find it, and it controls what kind of storage services you use and how your data is protected.

Storage accounts

- Must have a globally unique name.
- Provide over-the-internet access worldwide.
- Determine storage services and redundancy options.



Storing data once is risky. What if the server crashes? Azure solves this by keeping multiple copies of your data in different locations. This is called redundancy. The table below shows the four levels of protection Azure offers, from keeping copies within one building (LRS) all the way to spreading them across multiple regions and zones (GZRS). The more copies, the higher the durability rating.

Storage redundancy

Redundancy configuration	Deployment	Durability
Locally redundant storage (LRS)	Single datacenter in the primary region	11 nines
Zone-redundant storage (ZRS)	Three availability zones in the primary region	12 nines
Geo-redundant storage (GRS)	Single datacenter in the primary and secondary region	16 nines
Geo-zone-redundant-storage (GZRS)	Three availability zones in the primary region and a single datacenter in the secondary region	16 nines

Not all data is the same. A video file, a virtual machine disk, a chat message, and a spreadsheet all need to be stored differently. Azure offers five specialized storage services to handle each data type the right way, from unstructured text and binary data to virtual machine disks, queued messages, shared network files, and structured table data.

Azure storage services



Azure Blob: Optimized for storing massive amounts of unstructured data, such as text or binary data.



Azure Disk: Provides disks for virtual machines, applications, and other services to access and use.



Azure Queue: Message storage service that provides storage and retrieval for large amounts of messages, each up to 64 KB.



Azure Files: Sets up a highly available network file share that can be accessed by using the Server Message Block protocol.



Azure Tables: Provides a key/attribute option for structured nonrelational data storage with a schema-less design.

Storing data costs money, so Azure lets you choose how much you pay based on how often you actually need to access it. Data you open every day belongs in one tier, while files you might only need once a year belong in another. The four tiers below go from Hot (frequent access) all the way down to Archive (rarely touched), helping you cut storage costs without deleting anything.

Azure storage access tiers

Hot	Cool	Cold	Archive
Optimized for storing data that is accessed frequently.	Optimized for storing data that is infrequently accessed and stored for at least 30 days.	Optimized for storing data that is infrequently accessed and stored for at least 90 days.	Optimized for storing data that is rarely accessed and stored for at least 180 days with flexible latency requirements.

Every storage service in Azure has a web address, just like a website has a URL. This address is called a public endpoint and it is how your applications, users, or scripts talk to your storage over the internet. The table below shows the specific web address format for each of the five Azure storage services.

Storage service public endpoints

Storage service	Public endpoint
Blob Storage	https://<storage-account-name>.blob.core.windows.net
Data Lake Storage Gen2	https://<storage-account-name>.dfs.core.windows.net
Azure Files	https://<storage-account-name>.file.core.windows.net
Queue Storage	https://<storage-account-name>.queue.core.windows.net
Table Storage	https://<storage-account-name>.table.core.windows.net

Moving an entire business from physical servers to the cloud is not a small task. Azure Migrate exists to make that transition easier. It acts as a central hub where you can assess your current setup, figure out what needs to move, and then carry out the migration in an organized way. It brings together both Microsoft tools and third party tools under one roof.

Azure Migrate

- Unified migration platform.
- Range of integrated and standalone tools.
- Assessment and migration.



What do you do when your data is too massive to upload over the internet or when your location has little to no internet connection? Azure sends you a physical device called a Data Box. You load up to 80 terabytes of data onto it, ship it back to Microsoft, and they upload it directly into the cloud. It is also useful for moving data out of Azure for legal or compliance reasons.

Azure Data Box

- Store up to 80 terabytes of data.
- Move your disaster recovery backups to Azure.
- Protect your data in a rugged case during transit.
- Migrate data out of Azure for compliance or regulatory needs.
- Migrate data to Azure from remote locations with limited or no connectivity.



Once your data is in Azure, you need ways to manage and move files around. Azure gives you three tools depending on how comfortable you are with technology. AzCopy is for those who prefer typing commands. Azure Storage Explorer gives you a visual desktop interface like Windows Explorer. And Azure File Sync keeps your local files and cloud files in sync automatically, so you always have the latest version everywhere.

File management options

AzCopy

- Command-line utility.
- Copy blobs or files to or from your storage account.
- One-direction synchronization.

Azure Storage Explorer

- Graphical user interface (similar to Windows Explorer).
- Compatible with Windows, MacOS, and Linux.
- Uses AzCopy to handle file operations.

Azure File Sync

- Synchronizes Azure and on-premises files in a bidirectional manner.
- Cloud tiering keeps frequently accessed files local, while freeing up space.
- Rapid reprovisioning of failed local server (install and resync).

Identity, Access and Security

Now that your data is stored in the cloud, who gets to access it? That is where identity management comes in. Microsoft Entra ID is Azure's cloud service for verifying who you are and managing what you can do. It handles employee logins, allows one password to work across multiple apps through Single Sign On, manages applications, and even connects businesses with each other through B2B access.

Microsoft Entra ID

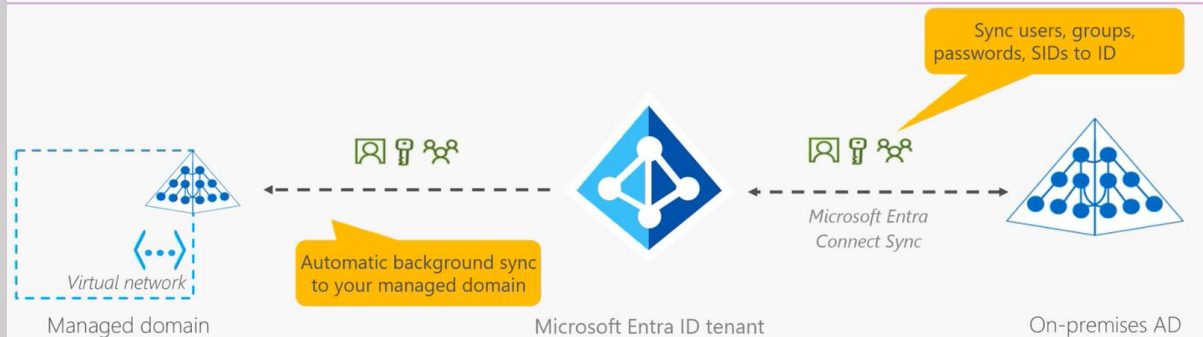
Microsoft Entra ID is Microsoft Azure's cloud-based identity and access management service.

- Authentication (employees sign in to access resources).
- Single sign-on (SSO).
- Application management.
- Business to Business (B2B).
- Device management.



Some older company applications were built before the cloud even existed, and they depend on traditional on-premises systems to work. Microsoft Entra Domain Services bridges that gap. It lets you bring those old applications into the cloud without changing them, while still syncing all your users, passwords, and group settings automatically from your existing directory.

Microsoft Entra Domain Services



- Gain the benefit of cloud-based domain services without managing domain controllers.
- Run legacy applications (that can't use modern auth standards) in the cloud.
- Automatically sync from Microsoft Entra ID.

Two words that sound similar but mean very different things in security. Authentication is about proving who you are, like showing your ID at the door. Authorization is about what you are allowed to do once you are inside, like which rooms you can enter. The slide below breaks down the key differences between both concepts using simple examples.

Compare authentication and authorization

Authentication

- Identifies the person or service seeking access to a resource.
- Requests legitimate access credentials.
- Basis for creating secure identity and access control principles.



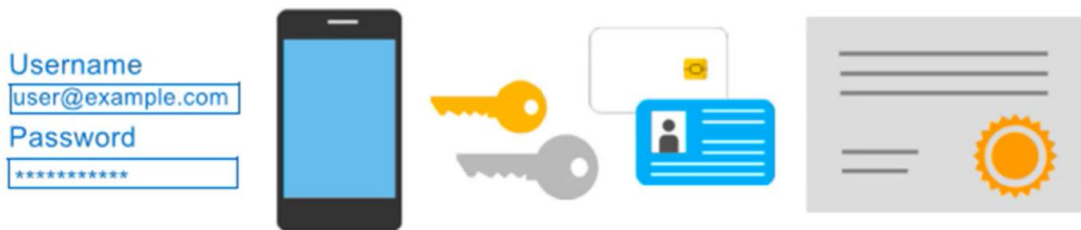
Authorization

- Determines an authenticated person's or service's level of access.
- Defines which data they can access, and what they can do with it.



A password alone is no longer enough to keep an account safe. Multifactor authentication adds extra layers of proof. Even if someone steals your password, they still cannot get in without also having your phone, your fingerprint, or another form of verification. MFA combines at least two of these three factors: something you know, something you physically have, and something that is uniquely you.

Multifactor authentication



Provides additional security for your identities by requiring two or more elements for full authentication.

- Something you know $\leftrightarrow$ Something you possess $\leftrightarrow$ Something you are